Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. **Participant Name (Team Name):** Inference Engines

b. **Participants:** Deeptesha Satpute, Somi Jeong, Lavanya Puri, Vedant Chaudhari

c. **Problem Statement:** Build a multi-agent AI system that helps users manage tasks, schedules, and information by interacting with multiple tools and data sources.
CareFlow - Multi-Agent Healthcare AI System

The Idea

CareFlow - Healthcare Post-Visit Care Coordinator

"Closing the gap between 'Doctor said' and 'Patient did"

Problem: Over 537 million adults globally live with chronic diseases. After a hospital visit, the care system often breaks down: 50% of patients forget medication instructions within 48 hours, caregivers remain in the dark, and doctors lack real-time data for follow-ups.

With One Voice/Text Input, CareFlow:

- ➔ Tracks medication schedules
- ➔ Books follow-up tests & appointments
- ➔ Stores medical visit records
- ➔ Notifies remote caregivers
- ➔ Proactively suggests health insights

The Gap It Fills

"The gap in care begins the moment the patient leaves the doctor's office."

Before CareFlow: 15-30 min manual work + high failure rate
After CareFlow: 10 seconds, AI handles everything

The Mission: To provide high-quality, proactive care coordination for elderly patients (like "Rajesh") and peace of mind for remote caregivers (like "Priya").

How We Built It

Technical Approach - ADK + MCP + AlloyDB

Use of ADK: Root Agent (Gemini 2.5 Pro) routes to 8 sub-agents. Parallel + sequential workflows

Use of MCP: Calendar MCP (appointments), Gmail MCP (caregivers), MCP Toolbox (DB queries)

Use of AlloyDB: Patient records + PG Vector for semantic search. No separate vector DB needed

Real-World Problem: India: 43.4% diabetes medication adherence. $100-300B annual global cost

Core WorkFlow: Voice input → Boss Agent → Task || Schedule (parallel) → Medical Info → Caregiver (sequential)

3 users connected: Patient (Rajesh, 63) → Caregiver (Priya, daughter in Bangalore) → Doctor (Dr. Patel)

Multi-Agent Orchestration: Powered by the **Google ADK**, we built a "Root Agent" that acts as a manager, delegating tasks across **8 specialized sub-agents** using advanced patterns:
Parallel Execution, Sequential Execution and **Loop Execution**

What Makes Us Different?

USP - From Assistive to Autonomous

Dimension	Typical Assistant	CareFlow
Role	Responds when asked	Proactively suggests insights
User	Patient only	Patient → Caregiver → Doctor
Action	Gives advice	Takes action, not just notes
Input	Text only	Voice + Text (for elderly patients)

Patient speaks. Caregiver knows. Doctor is prepared.

What others miss:

- ➔ Most are schedule managers, not healthcare-specific
- ➔ Most don't connect caregivers
- ➔ Most wait for you to ask - CareFlow speaks first
- ➔ **Fail-Safe Guardrails:** 2-layer safety architecture that hard-blocks medical errors like dosage or allergy risks.

Opportunity:

- ➔ Aging population + chronic disease = global crisis
- ➔ India resonates
- ➔ This scales
- ➔ **Proactive & Clinical:** Detects health trends (BP/Glucose) automatically using precise, clinically-tuned AI models.

Core Features

What CareFlow Provides

Patient-Facing

Voice/Text Intake
Doctor visit auto-processed

Medication Tracking
Extract, update & remind

Smart Appointment Booking
Checks calendar, books fasting labs

Medical Record RAG
Semantic search over past visits

Health Trend Analysis
Proactive BP & vitals insights

Caregiver-Facing

Visit Update Notifications
Immediate email post-visit

Weekly Health Digest
Adherence + appointments

Emergency Alerts
Missed meds + abnormal symptoms

Medical Staff

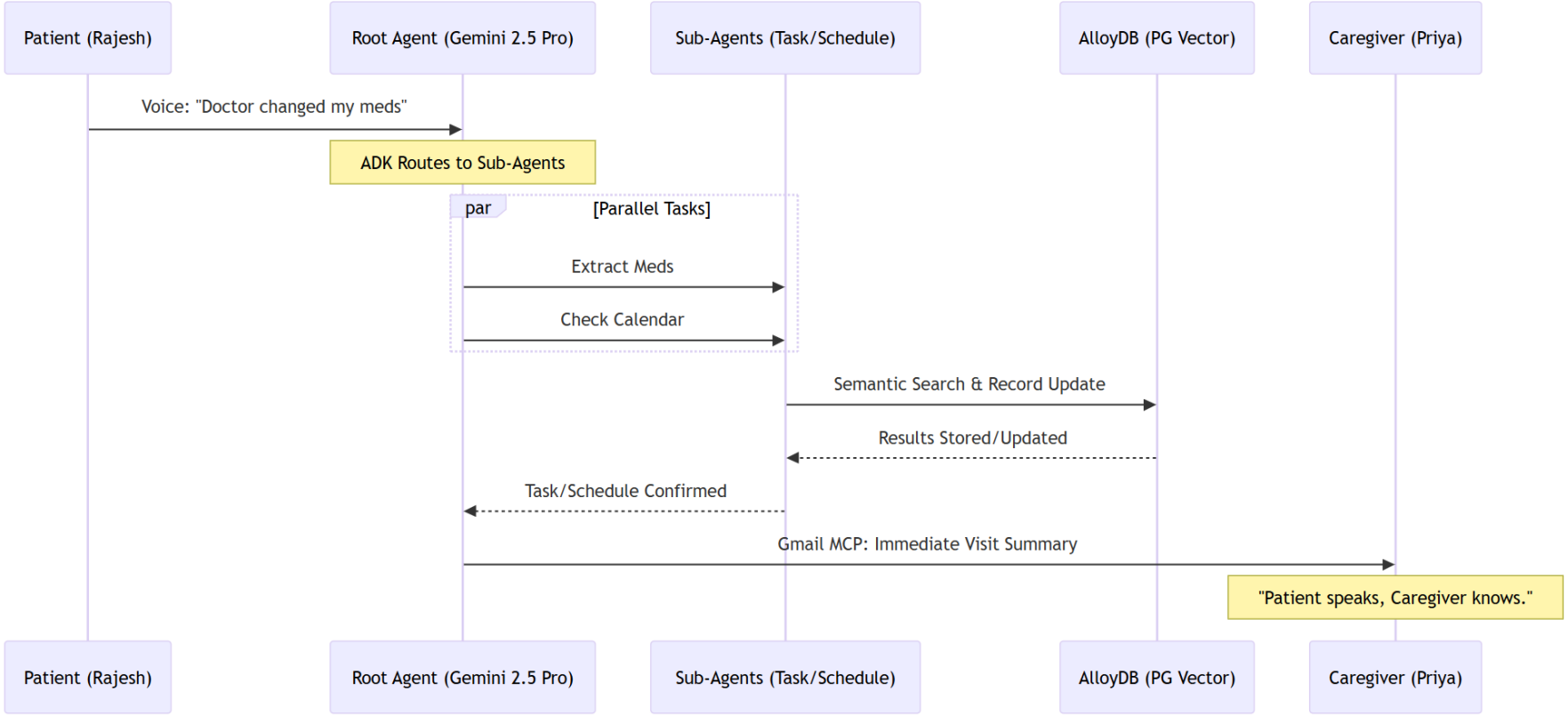
Pre-Visit Summary
Trends, adherence & concerns

Diet Guidance
Meal recs based on doctor notes

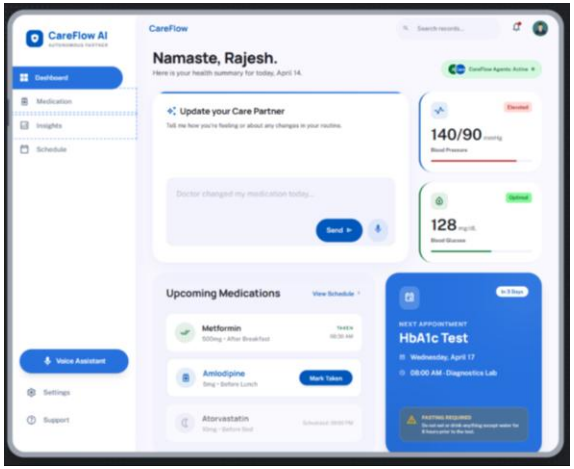
Symptom Triage
Urgency classification + escalation

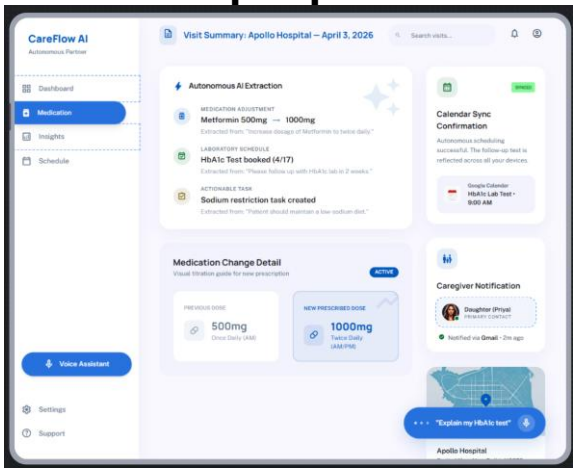
Process Flow Diagram

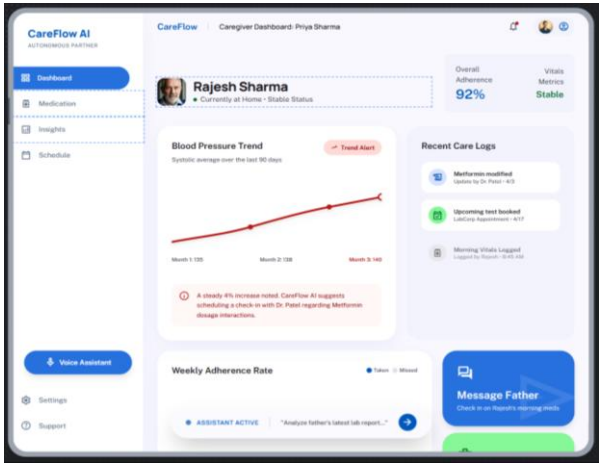
How CareFlow Works - One Input, Eight Agents, Three Users

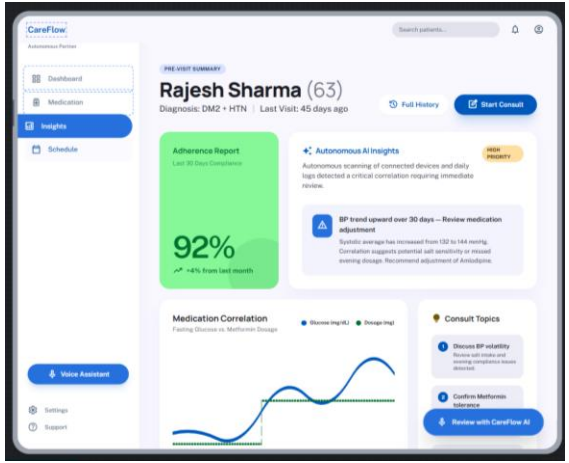


Wireframes/Mock diagrams of the proposed solution





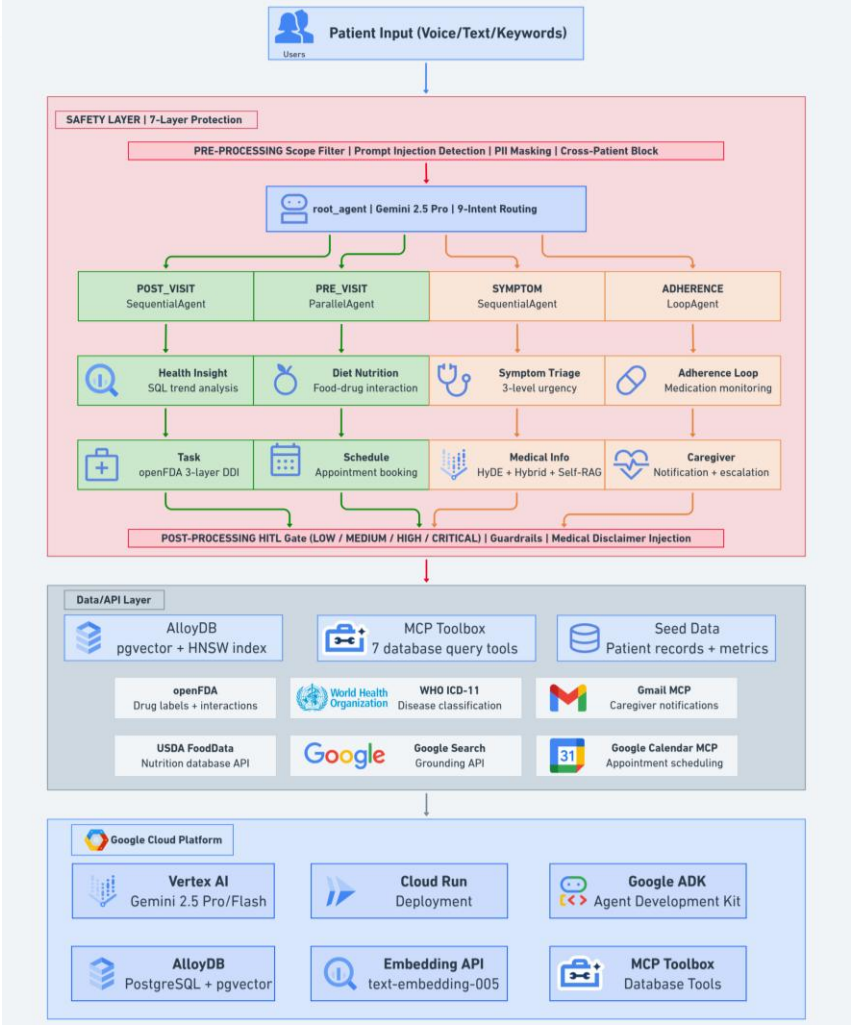




Architecture Diagram

The Engine Behind CareFlow

- **8-Agent Orchestration:** Utilizes Parallel, Sequential, and Loop ADK patterns to isolate clinical workflows, preventing monolithic prompt confusion.
- **5-Stage Agentic RAG:** Replaces vanilla RAG with a pipeline featuring HyDE, Hybrid Search (pgvector Dense + BM25), Reciprocal Rank Fusion (RRF), and Self-RAG reflection.
  Deterministic Safety Layer: Implements a 2-layer scope filter and hardcoded clinical guardrails (e.g., maximum dosages) that the LLM cannot override.
- **Custom SSE MCP Server:** Uses Server-Sent Events for state-aware Gmail and Google Calendar tool integration with autonomous OAuth token management.
- **4-Level HITL Escalation:** Context-aware Human-in-the-Loop matrix that automatically halts execution and demands user confirmation for high-risk actions.
- **AlloyDB + pgvector Core:** Consolidates millisecond-latency transactional patient data and 768-dimensional vector embeddings into a single infrastructure.



Technologies/Google Services used in the solution

The ”Root Agent" (Gemini 2.5 Pro) now acts as a high-speed dispatcher, using **ADK** to manage multiple sub-agents and tools simultaneously.

- **The Brain (Gemini 2.5 Pro):** Processes voice/text inputs and determines the urgency of the task.
- **The Memory (AlloyDB + PG Vector):** Stores patient history and contact details for Rajesh, Priya and Dr. Patel. Secures the clinical history of Rajesh and his family.

  Uses **medical embeddings** (text-embedding-005) for 100% factual RAG retrieval.
- **The Hands (MCP + Twilio)**:

  Gmail MCP: Sends detailed visit records, weekly health digests and used for alerts on missed medications, etc.

  Calendar MCP: Automatically books follow-up tests and appointments.

  Twilio API: Triggers immediate SMS/WhatsApp alerts for missed medications or critical symptoms

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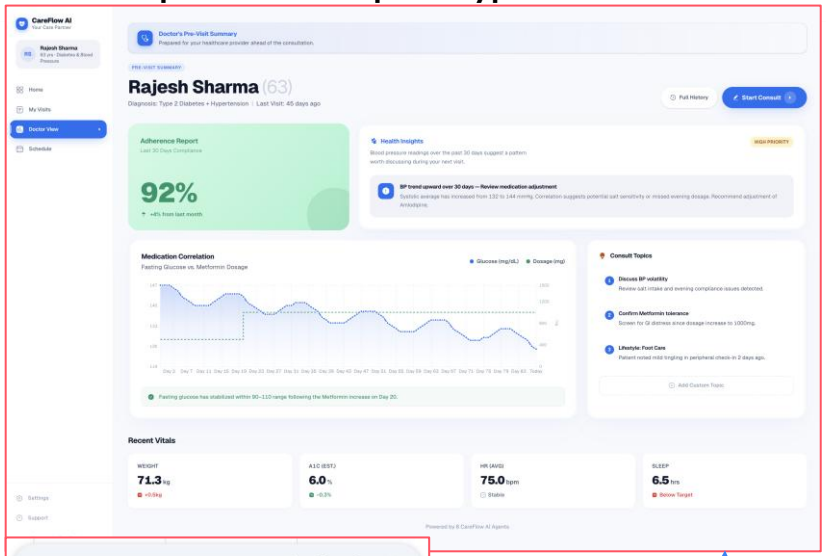
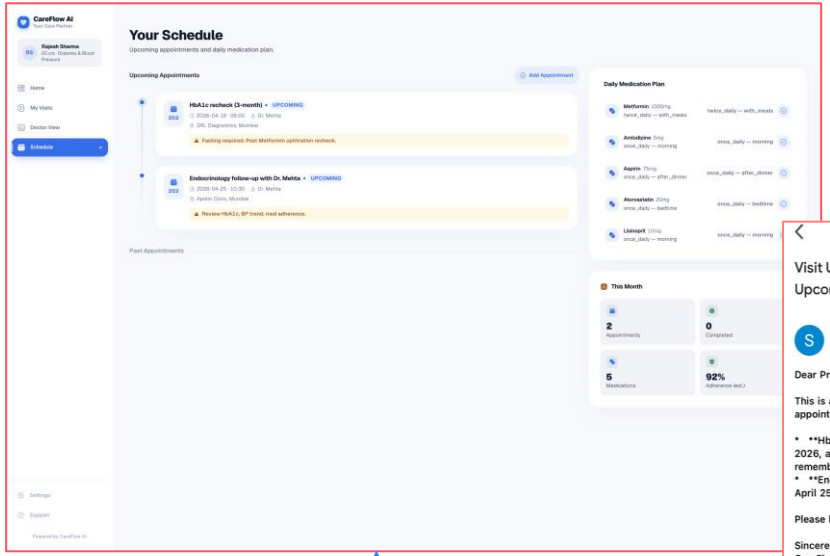
Snapshots of the prototype

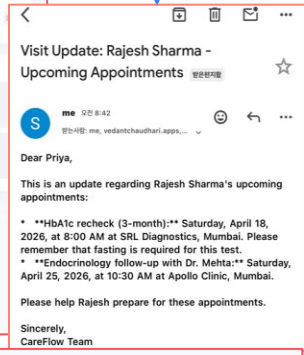


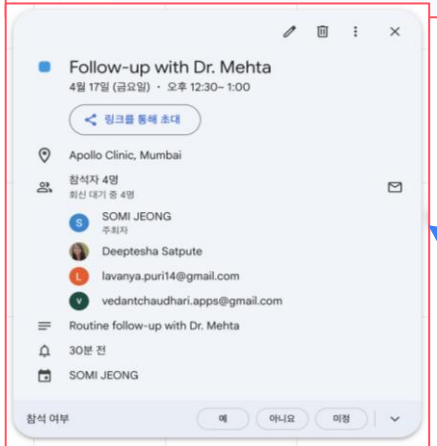
Doctor View for all Vitals



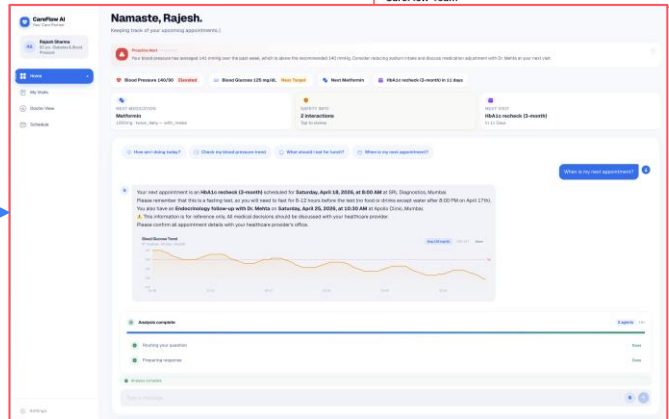
Schedule View for Patient showing them their next appointments and Daily Medication Plans

Gmail Updates Received by Caretaker using Gmail MCP





Google Calendar MCP in action Adding Events on Calendar for Patient and Caretaker



Home View for Patient where they can ask questions to agent. Given snapshot shows Agentic RAG in action

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Thank you